



United International University
Department of Computer Science and Engineering
Course Code: CSE 2215 Course Title: Data Structure and
Algorithms I (B)

Class Test – 5
Duration: 30mins (all inclusive)

Date: 28/09/2021
Marks: 20

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1. Construct a **Binary Max Heap** from the given sequence of numbers after your student id in the following table. Draw a separate tree for each iteration of the process. [15]
 2. Then apply these operations in the order they are given. [5]
 - i. `extract_max()`
 - ii. `insert(35)`

Student ID	Given array
011173044	[11, 17, 20, 22, 31, 35, 39, 41, 42, 45]
011181265	[13, 21, 24, 29, 30, 32, 34, 35, 39, 40]
011183017	[10, 13, 17, 18, 22, 25, 34, 35, 39, 48]
011183020	[13, 14, 15, 19, 24, 33, 37, 39, 41, 42]
011183056	[13, 15, 17, 34, 37, 39, 42, 44, 47, 49]
011183077	[10, 12, 16, 18, 23, 30, 32, 36, 46, 48]
011191160	[11, 12, 13, 18, 25, 29, 31, 34, 40, 49]
011191173	[13, 17, 24, 25, 26, 28, 30, 33, 43, 49]
011191179	[16, 21, 23, 28, 29, 31, 41, 44, 45, 46]
011191212	[15, 22, 25, 27, 30, 33, 34, 37, 38, 42]
011192083	[10, 24, 25, 27, 32, 38, 39, 41, 45, 46]
011192094	[12, 13, 17, 22, 23, 31, 40, 42, 44, 48]
011192122	[13, 24, 25, 26, 27, 34, 35, 41, 42, 45]
011193028	[15, 21, 24, 25, 28, 30, 34, 35, 43, 48]
011193038	[11, 24, 25, 26, 29, 30, 34, 38, 40, 49]
011193040	[11, 15, 16, 21, 22, 23, 24, 29, 36, 48]
011193045	[18, 22, 23, 26, 28, 29, 31, 33, 41, 46]
011193047	[17, 18, 22, 25, 26, 30, 31, 34, 44, 45]
011193052	[10, 14, 21, 22, 23, 24, 27, 28, 32, 49]
011193068	[10, 11, 12, 17, 21, 29, 31, 38, 40, 47]

011193122	[13, 14, 20, 23, 26, 31, 38, 44, 45, 49]
011193123	[12, 15, 24, 28, 31, 34, 35, 36, 40, 44]
011201019	[13, 14, 17, 30, 31, 35, 36, 42, 47, 49]
011201031	[12, 14, 19, 22, 34, 38, 40, 43, 46, 49]
011201037	[11, 13, 14, 21, 25, 35, 42, 45, 47, 49]
011201084	[11, 30, 31, 34, 41, 43, 44, 47, 48, 49]
011201088	[14, 19, 26, 32, 33, 34, 35, 36, 42, 43]
011201108	[14, 15, 17, 20, 26, 28, 40, 46, 47, 49]
011201125	[13, 14, 24, 34, 35, 36, 42, 45, 46, 48]
011201128	[12, 19, 24, 25, 26, 34, 35, 38, 41, 42]
011201156	[11, 12, 14, 21, 27, 31, 33, 40, 41, 46]
011201166	[11, 19, 24, 26, 28, 31, 36, 37, 47, 48]
011201170	[10, 13, 19, 33, 38, 39, 41, 43, 44, 47]
011201173	[11, 17, 28, 30, 34, 40, 42, 43, 44, 48]
011201176	[17, 21, 25, 26, 27, 28, 29, 42, 44, 45]
011201286	[17, 18, 26, 28, 30, 32, 39, 44, 46, 49]
011201334	[11, 12, 24, 25, 27, 34, 38, 40, 43, 46]
011201344	[11, 13, 20, 21, 22, 28, 31, 33, 43, 44]
011201352	[10, 11, 17, 22, 27, 32, 33, 39, 41, 46]
011201353	[14, 19, 24, 26, 31, 34, 37, 42, 46, 48]
011201366	[14, 17, 27, 31, 33, 37, 41, 46, 47, 49]
011201400	[10, 16, 21, 22, 23, 24, 30, 34, 44, 49]
011201403	[16, 17, 20, 21, 24, 25, 30, 42, 44, 45]
011201406	[14, 16, 17, 19, 22, 23, 35, 37, 39, 41]
011201446	[10, 12, 20, 23, 27, 28, 29, 34, 41, 49]
011202199	[11, 13, 14, 19, 28, 29, 32, 35, 46, 49]